

4MRNA User Manual

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1. Introduction

1-1. What is 4MRNA?

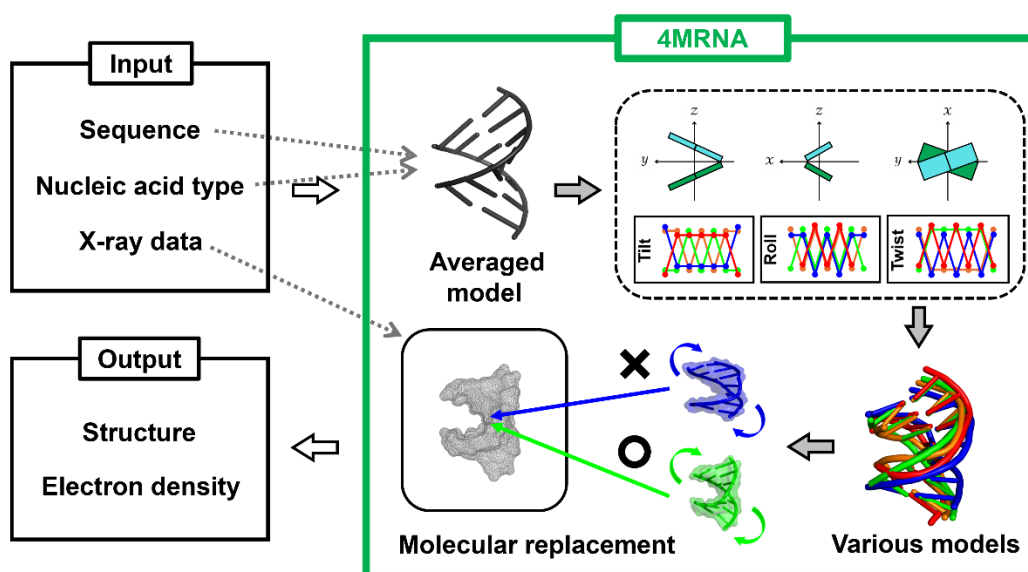
4MRNA is an abbreviation for “**M**assive **M**ulti-type **M**odel **M**olecular **R**eplacement for **N**ucleic **A**acids”. This is a novel method designed to enhance molecular replacement (MR) for phasing in X-ray crystallography of nucleic acids.

Applying MR, which is widely used in structure determination (phase determination), to nucleic acids presents unique challenges that are not encountered with proteins. To overcome these issues, we developed an innovative strategy termed 4MRNA.

One major difficulty is that nucleic acids can adopt different 3D structures even with the same sequence. As a result, models existing in the database may not be suitable as search models, and sequence-based structure prediction methods such as AlphaFold are also limited in applicability. Moreover, it is empirically known that MR of nucleic acids can fail even when the search model and the target structure differ only slightly. To address these issues, the new strategy 4MRNA includes creating a large number of diverse search models (= massive multi-type models) and applying them to MR.

We found that by varying three out of the twelve parameters that control the three-dimensional structures of nucleic acids according to different patterns, the resulting set of models included ones that closely matched the correct structure, thereby increasing the success rate of MR.

Building on this strategy, we employed 3DNA, which is a software that generates nucleic acid structural models based on parameters, to create a wide variety of models. The processes of parameter adjustment and model creation have been automated using Shell scripts. Subsequently, MR is carried out for each of the many created models. Since this operation must be repeated many times, we also automated this process using Shell scripts.



1-2. Two modes to execute 4MRNA

There are two modes to execute 4MRNA: **default mode** and **customize mode**.

For both mode, you need to install the **4MRNA** command on your computer in advance.

In the **default mode**, you can simply type the command **4MRNA** in Ubuntu/Terminal and run it. Thus, it is well suited for all users.

In the **customize mode**, you can obtain some programming codes by running the command **4MRNA**. Basically, you just run them in the prescribed order, but if needed, you can edit and customize the code. For example, you can adjust parameters by yourself to get more diverse models, or add models prepared by other methods to perform molecular replacement. Such customization represents a highly advanced option and is generally intended for experts in structural biology or X-ray crystallography.

Style	Procedure	Pre-setup	Usability	Customization	Target users
Default	Run the command 4MRNA	Required	Very good	Not possible	For all users
Customize	Download codes → Run them in order	Required	Fair	Possible	For experts

For details on how to make the preparations, please refer to the Preparations section.

⇒ [Preparations](#)

For details on the usage of each mode, please refer to the corresponding sections.

⇒ [Default mode](#)

⇒ [Customize mode](#)

2. Preparations

The software “4MRNA” consists of shell scripts, so and can therefore run on Windows via WSL (using a distribution such as Ubuntu), as well as on macOS and Linux.

2-1. How to download and install 4MRNA

Please run the following commands in order.

```
cd $HOME
git clone https://github.com/S-Ando-Biophysics/4MRNA-Install.git
bash 4MRNA-Install/install.sh
echo 'export PATH="$HOME/4MRNA-Install/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

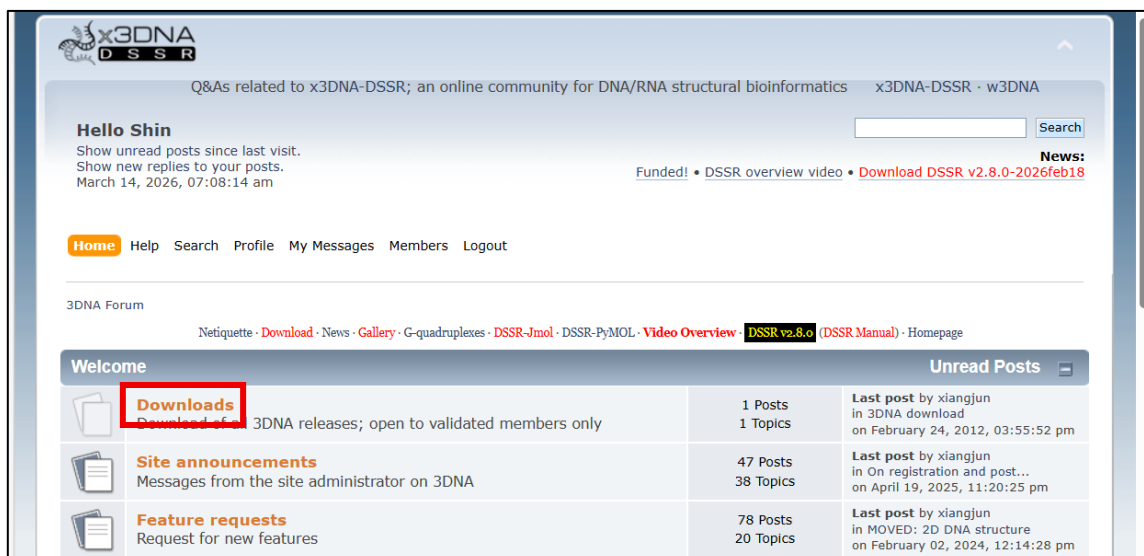
2-2. Additional preparation

Please install and set up the following softwares in advance.

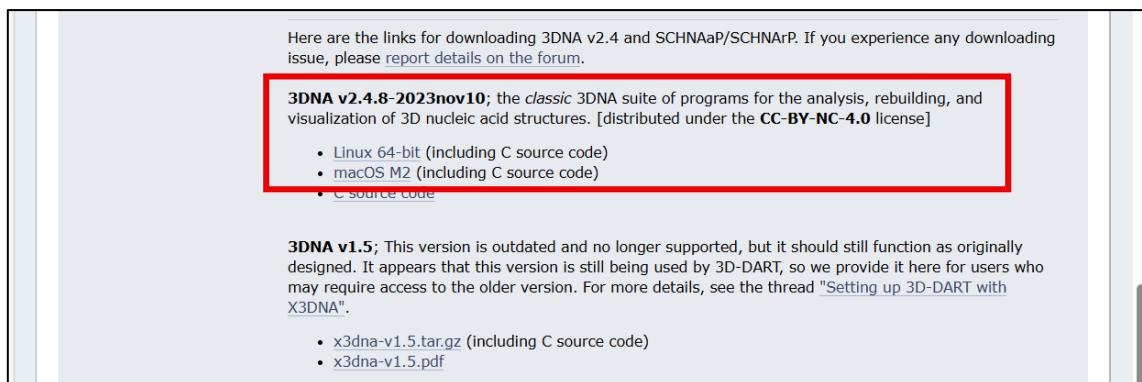
Name	Download link	Remarks
3DNA	http://forum.x3dna.org	It can be downloaded after registering on the site and receiving approval.
Phenix	https://phenix-online.org/download	The command-line version is required.
Phaser	https://www.ccp4.ac.uk/download	The program Phaser is included in CCP4, so please install CCP4.

2-2-1. 3DNA

Step 1. Visit the [official website](http://forum.x3dna.org) and register on the site. Once you receive the approval email a few days later, you will be able to download 3DNA.



Step 2. Download the installer for the Linux or macOS version. Choose the Linux version even if you're using Windows.

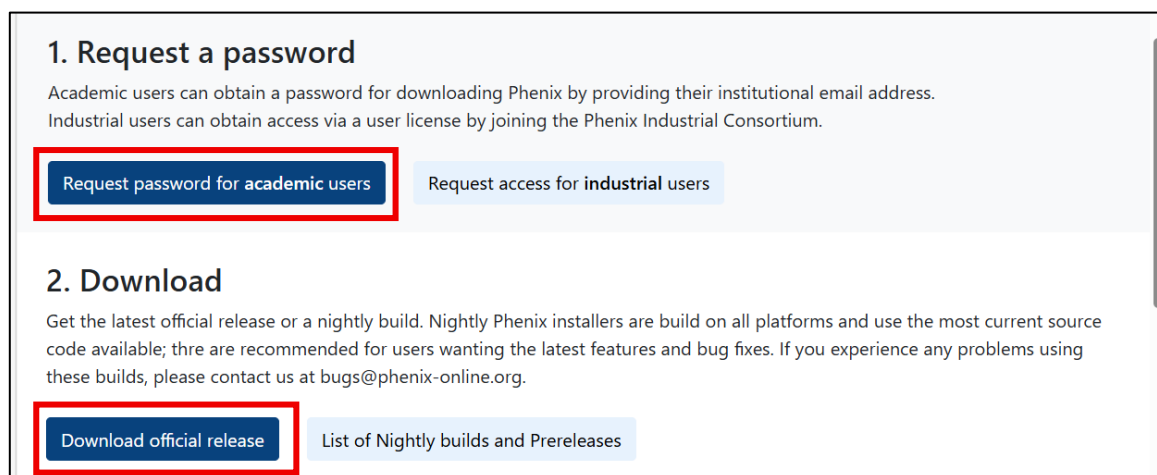


Step 3. Once the download is complete, run the following commands in order to complete the installation. The following commands are for Ubuntu on Windows, but similar commands should be used for macOS and Linux. (These commands assume that the file “x3dna-v2.4-linux-64bit.tar.gz” has been downloaded to “C:\Users\name\Downloads”. You will need to adjust the version number and directory path as appropriate.)

```
sudo su
cd /usr/local
mv /mnt/c/Users/name/Downloads/x3dna-v2.4-linux-64bit.tar.gz .
tar -pzxvf x3dna-v2.4-linux-64bit.tar.gz
cd x3dna-v2.4/bin
./x3dna_setup
exit
echo 'export X3DNA=/usr/local/x3dna-v2.4' >> ~/.bashrc
echo 'export PATH="$X3DNA/bin:$PATH"' >> ~/.bashrc
source ~/.bashrc
```

2-2-2. Phenix

Step 1. Visit the [official website](#) and request a password. Once you receive the email a few minutes later, you will be able to download Phenix.



Step 2. Download the command-line installer for the Linux or macOS version. Choose the Linux version even if you're using Windows.



The screenshot shows a web page with two main sections: Linux and macOS. The Linux section has a purple header and lists two options: 'x86-64' and 'CUDA 12 x86-64', both with a blue link to 'download command-line installer'. The macOS section also has a purple header and includes instructions for macOS 15 (Sequoia) and older versions, followed by a default installation path. It lists four options: 'Apple Silicon (macOS 11 and later)' with 'download GUI installer' and 'download command-line installer' links, and 'x86-64 (macOS 10.13 and later)' with 'download GUI installer' and 'download command-line installer' links. The 'download command-line installer' link for Apple Silicon is highlighted with a red box.

Linux

x86-64 [[download command-line installer](#)]

CUDA 12 x86-64 [[download command-line installer](#)]

macOS

For macOS 15 (Sequoia) and later, you **must** run `xattr -c <GUI installer name>` in the Terminal.app for macOS to run the GUI installer.

For older versions of macOS, you can right-click the GUI installer, and then select "Open" to skip the security error.

By default, the GUI installer will install Phenix in **/Applications** for all users or **~/Applications** for just yourself.

Apple Silicon (macOS 11 and later) [[download GUI installer](#)]

Apple Silicon (macOS 11 and later) [[download command-line installer](#)]

x86-64 (macOS 10.13 and later) [[download GUI installer](#)]

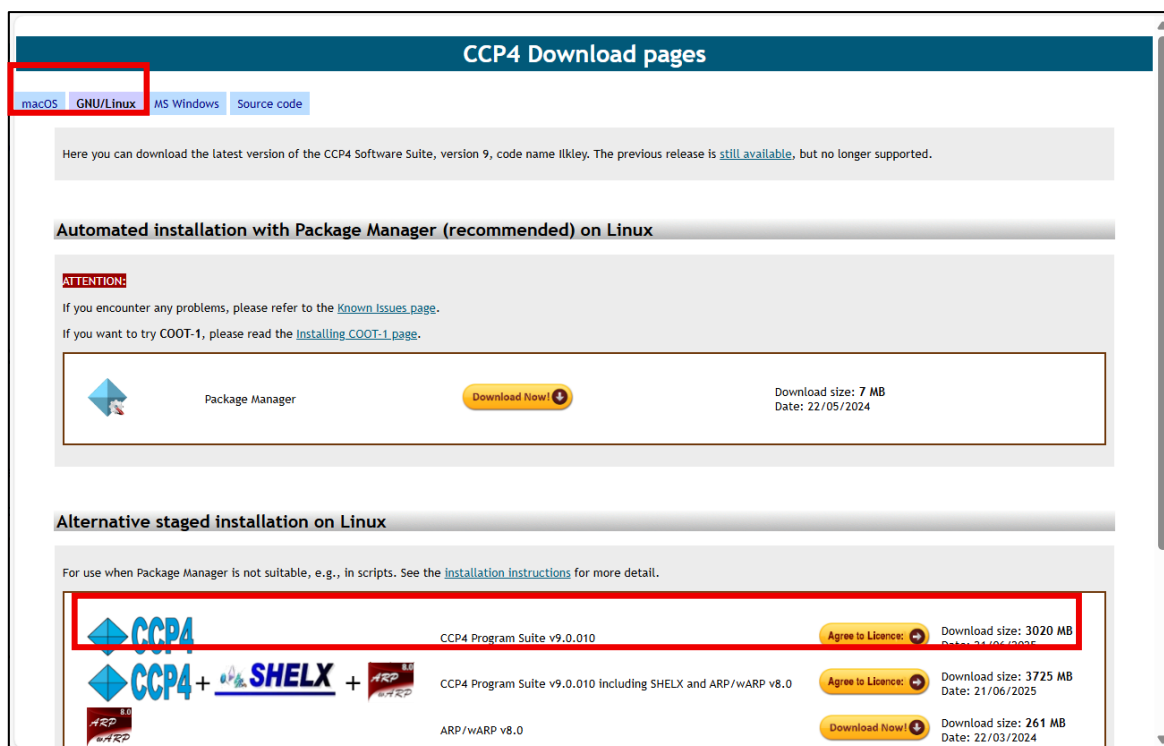
x86-64 (macOS 10.13 and later) [[download command-line installer](#)]

Step 3. Once the download is complete, run the following commands in order to complete the installation. The following commands are for Ubuntu on Windows, but similar commands should be used for macOS and Linux. (These commands assume that the file "Phenix-2.0-5936-Linux-x86_64.sh" has been downloaded to "C:\Users\name\Downloads". You will need to adjust the version number and directory path as appropriate.)

```
sudo su
cd /usr/local
mv /mnt/c/Users/name/Downloads/Phenix-2.0-5936-Linux-x86_64.sh .
bash Phenix-2.0-5936-Linux-x86_64.sh -b -p /usr/local/phenix-2.0-5936
exit
echo "source /usr/local/phenix-2.0-5936/phenix_env.sh" >> ~/.bashrc
source ~/.bashrc
```

2-2-3. Phaser

Step 1. Phaser is included in a software suite termed CCP4. Visit the [official website](#) and download the Linux or macOS version. Choose the Linux version even if you're using Windows.



Step 2. Once the download is complete, run the following commands in order to complete the installation. The following commands are for Ubuntu on Windows, but similar commands should be used for macOS and Linux. (These commands assume that the file “ccp4-9.0.010-linux64.tar.gz” has been downloaded to “C:\Users\name\Downloads”. You will need to adjust the version number and directory path as appropriate.)

```
sudo su
cd /usr/local
mv /mnt/c/Users/name/Downloads/ccp4-9.0.010-linux64.tar.gz .
gunzip ccp4-9.0.010-linux64.tar.gz
tar -xvf ccp4-9.0.010-linux64.tar
cd ccp4-9
./BINARY.setup
exit
echo "source /usr/local/ccp4-9/bin/ccp4.setup-sh" >> ~/.bashrc
source ~/.bashrc
```

3. How to use the **default mode**

In this mode, executing the **4MRNA** command automatically initiates the entire 4MRNA workflow, and you simply need to wait for it to complete.

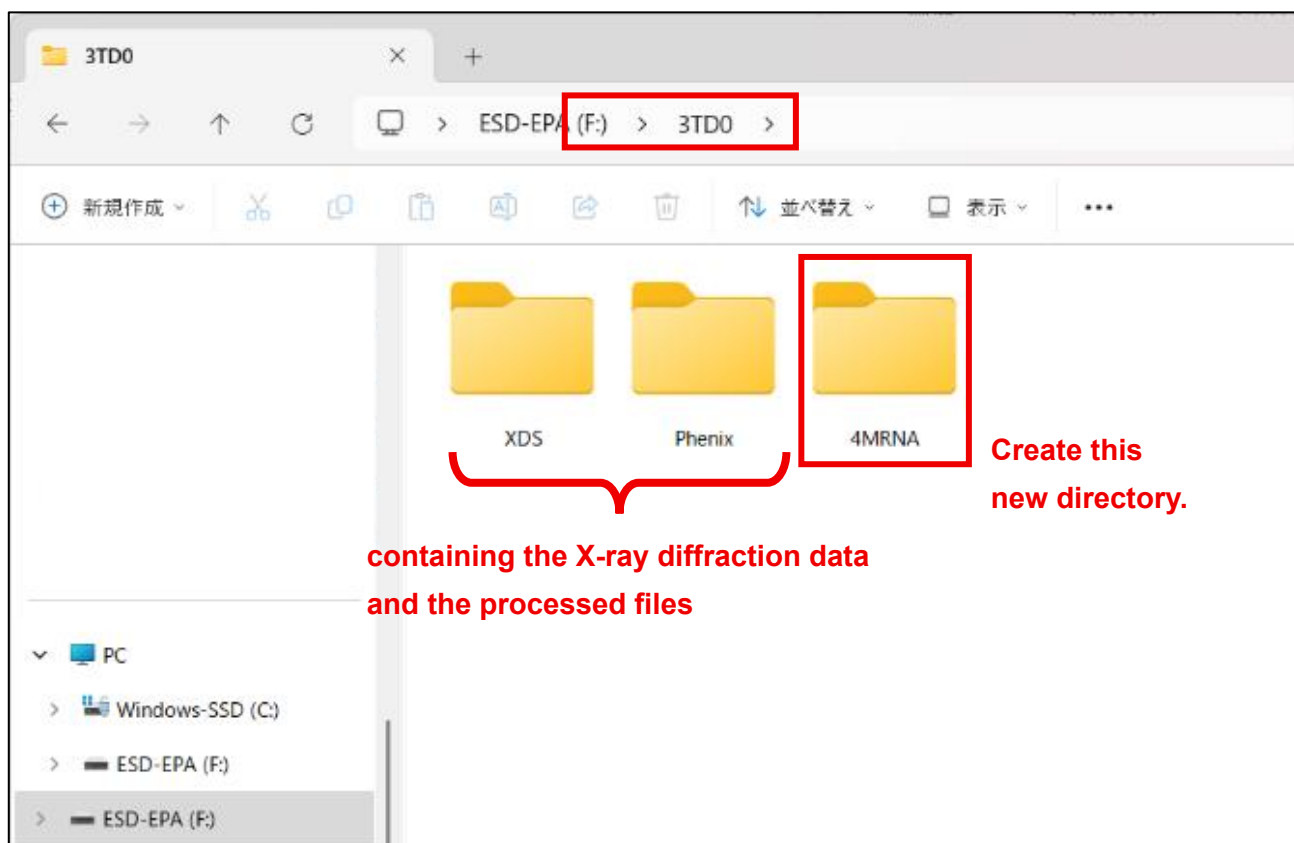
3-1. Summary of the procedure

- Step 1. Create a new directory on your computer for executing 4MRNA. After creating it, place your reflection file (.mtz) for molecular replacement into that directory.
- Step 2. Open Ubuntu/Terminal, change to the directory created in Step 1, and run the command **4MRNA**.
- Step 3. When prompted to choose the execution mode, select "**default**".
- Step 4. When 4MRNA finishes, a directory named "**4MRNA-Results**" will be created, containing up to seven candidate solutions for molecular replacement.

3-2. Step-by-step instructions with screenshots

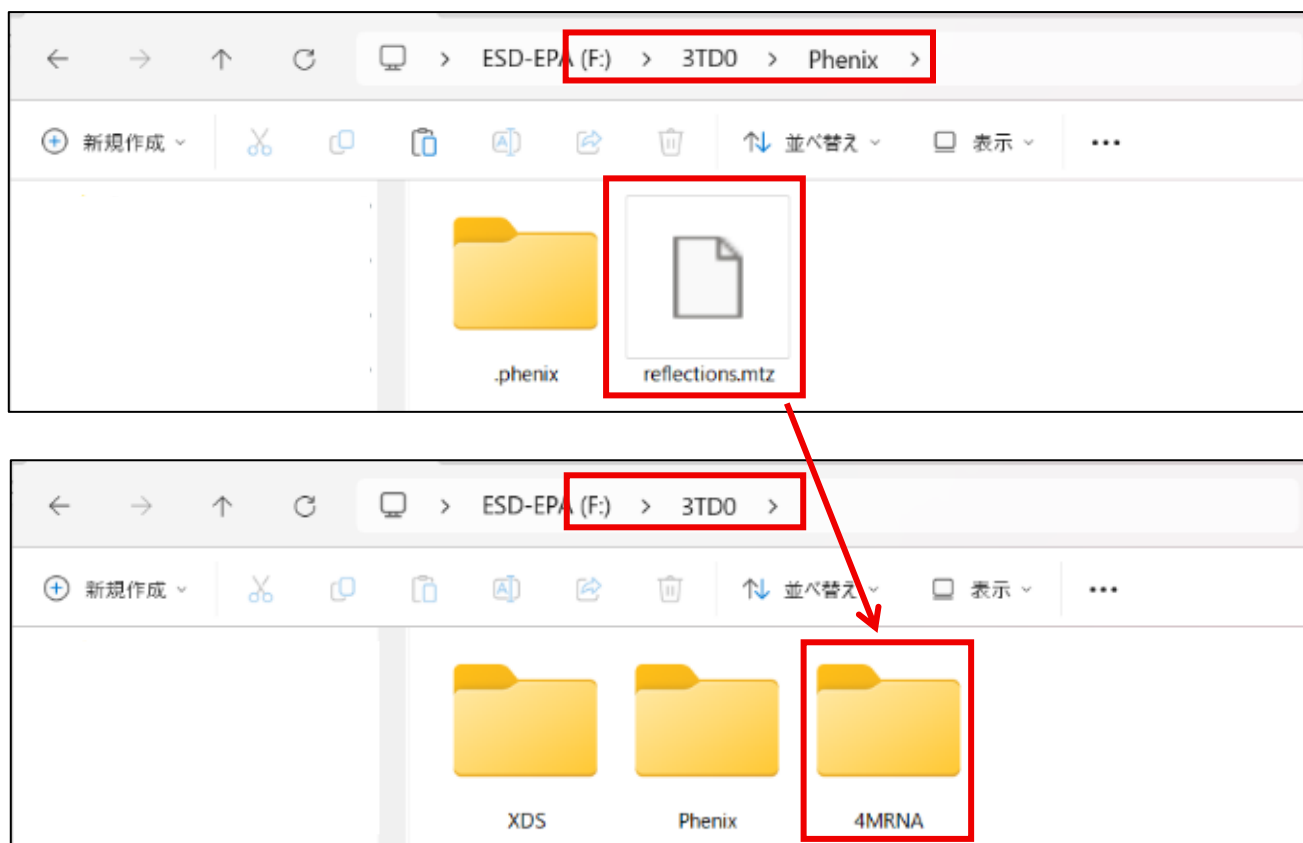
- Create a new directory on your computer for executing 4MRNA.

[e.g.] Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a directory 3TD0 on a USB drive (F:). In that case, create a new directory named "4MRNA" inside "F:\3TD0".

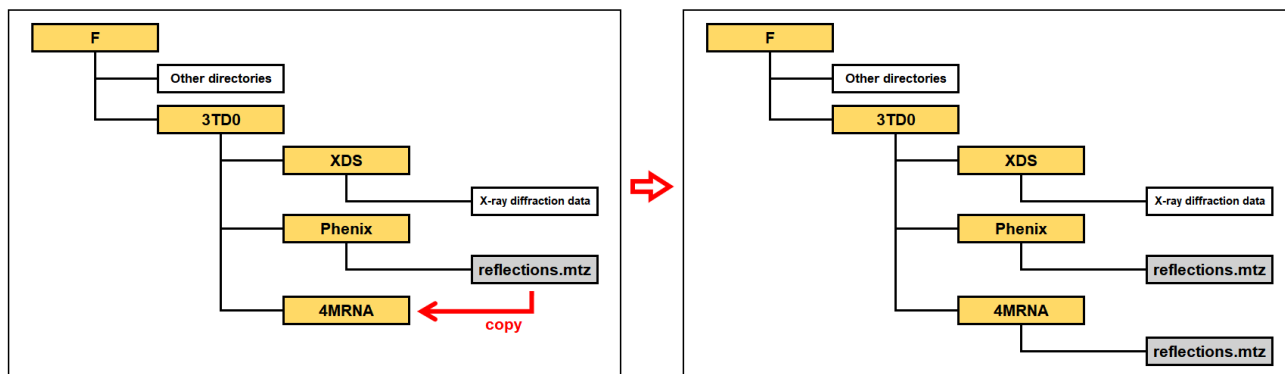


■ Place your reflection file (.mtz) for molecular replacement into the created directory.

[e.g.] When using the software Phenix, the reflection file is typically named “reflections.mtz”. If this file is in “F:\3TD0\Phenix”, copy it, and paste to “F:\3TD0\4MRNA”.



■ The directory structure after these operations is as illustrated in the example below.



■ Open Ubuntu or Terminal, change to the created directory, and run the command **4MRNA** as follows.

[e.g.] If the created directory is “F:\3TD0\4MRNA”, enter the command in Ubuntu as shown in the figure below. (This example is for Windows.)

```
shin@Shin-PC-Lenovo: /mnt/f. x + v
(base) shin@Shin-PC-Lenovo:~$ cd /mnt/f/3TD0/4MRNA
(base) shin@Shin-PC-Lenovo:/mnt/f/3TD0/4MRNA$ 4MRNA|
```

- Actually, for the command **4MRNA** to work, an input file called “**4MRNA-INP.txt**” is required. The user is prompted to choose from two options: (1) preparing the file yourself (a template will be provided), or (2) answering questions on the screen so that the program automatically generates it.
- Option (2) is strongly recommended. For option (1), refer to the supplementary notes in [Section 3-3](#).
- If you choose option (2), several questions will be asked. Please answer them. Once all questions have been answered, the 4MRNA workflow will start automatically.

[e.g.] The figure below shows an example of molecular replacement using two different RNAs as models.

In this example, during molecular replacement, one copy of the first model and two copies of the second model are searched for in the asymmetric unit of the crystal. In addition, the 5' end of a nucleic acid may retain a phosphate group or lack a phosphate group depending on the synthesis method. To allow a more accurate model to be used for molecular replacement, this can be specified in the model settings. (In this example, “Keep” option is selected with the refinement step after molecular replacement in mind.) This is a relatively minor detail and is often not critical. Therefore, if you are unsure, selecting “Keep” is generally a safe choice.

```
Do you want to run 4MRNA (v2.5.0) now? [Y/n] Y
[4MRNA] Information written to: /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Information.txt
[4MRNA] Downloading release v2.5.0...
Input file '4MRNA-INP.txt' not found.
Choose one of the following:
  1) I will prepare it myself. (A template will be provided.)
  2) I will answer questions here to generate it now.
Select [1/2]: 2
The number of models for molecular replacement [1,2,3]: 2

Enter information of model #1.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): CGUCGACG
Do you want to keep or remove the 5'-terminal phosphate group? [Keep,Remove]: Keep
How many copies of the model to be searched in molecular replacement: 1

Confirm model #1:
- TYPE1 A-RNA
- SEQ1 CGUCGACG
- 5PHOS1 Keep
- NUM1 1
Is this correct? [Y/N]: Y

Enter information of model #2.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): GCGUC
Do you want to keep or remove the 5'-terminal phosphate group? [Keep,Remove]: Keep
How many copies of the model to be searched in molecular replacement: 2

Confirm model #2:
- TYPE2 A-RNA
- SEQ2 GCGUC
- 5PHOS2 Keep
- NUM2 2
Is this correct? [Y/N]: Y
```

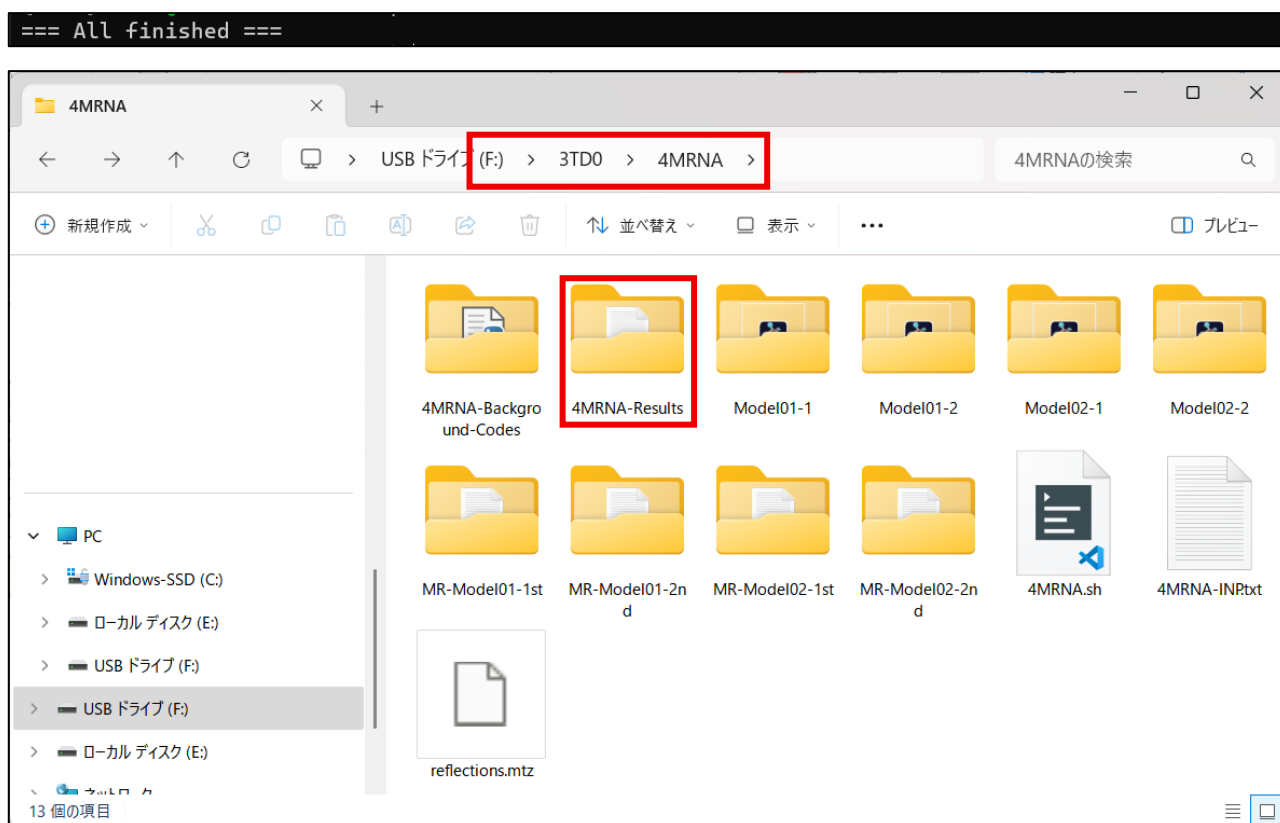
- When prompted with “Please choose execution mode.”, please enter “**default**”.

```
4MRNA-INP.txt created
- TYPE1 A-RNA
- SEQ1 CGUCGACG
- 5PHOS1 Keep
- NUM1 1

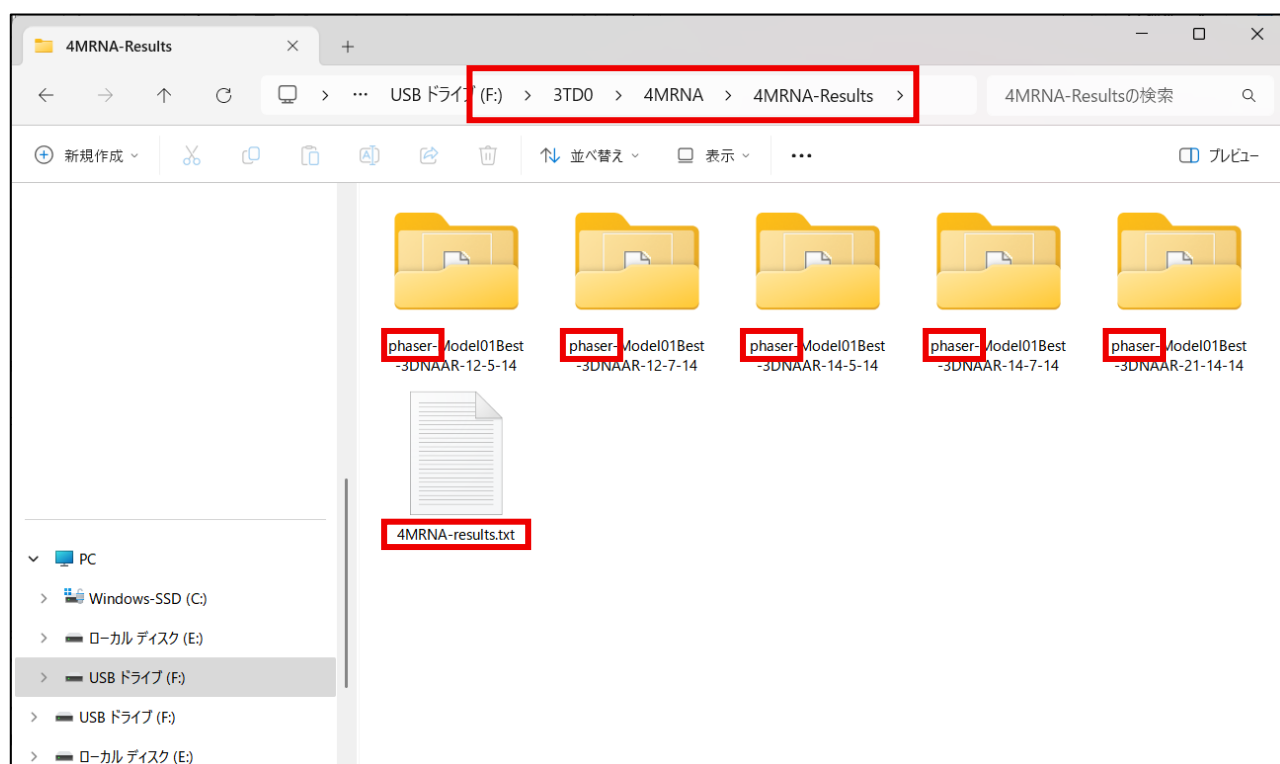
- TYPE2 A-RNA
- SEQ2 GCGUC
- 5PHOS2 Keep
- NUM2 2

Continuing...
Detected model types: 2
Model01: TYPE=A-RNA SEQ=CGUCGACG 5PHOS=Keep NUM=1 MW=5051.38
Model02: TYPE=A-RNA SEQ=GCGUC 5PHOS=Keep NUM=2 MW=3115.14
Downloading: 01.Shell-CreatingModels-1st.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/01.Shell-CreatingModels-1st.sh
Downloading: 02.Shell-MR-2model-Model01-1st.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/02.Shell-MR-2model-Model01-1st.sh
Downloading: 03.Shell-CreatingModels-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/03.Shell-CreatingModels-2nd.sh
Downloading: 04.Shell-MR-2model-Model01-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/04.Shell-MR-2model-Model01-2nd.sh
Downloading: 05.Shell-MR-2model-Model02-1st.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/05.Shell-MR-2model-Model02-1st.sh
Downloading: 06.Shell-CreatingModels-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/06.Shell-CreatingModels-2nd.sh
Downloading: 07.Shell-MR-2model-Model02-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/07.Shell-MR-2model-Model02-2nd.sh
Downloading: min.params -> /mnt/h/4MRNA-Update/v2.5.0/test3/min.params
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/01.Shell-CreatingModels-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/03.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/06.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/01.Shell-CreatingModels-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/02.Shell-MR-2model-Model01-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/03.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/04.Shell-MR-2model-Model01-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/05.Shell-MR-2model-Model02-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/06.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/07.Shell-MR-2model-Model02-2nd.sh
Please choose execution mode. [default,customize]: default
```

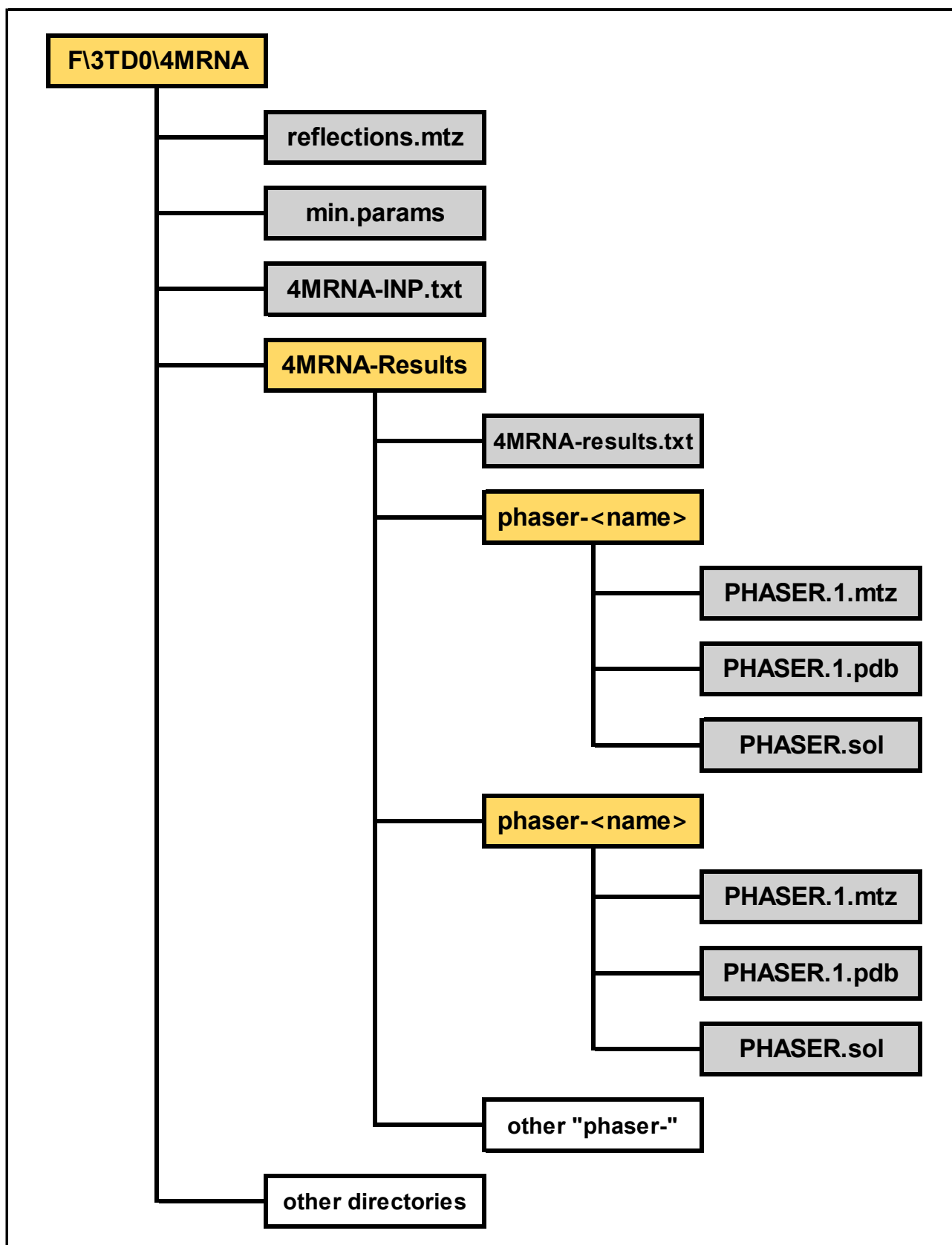
- When the entire workflow is finished, the message “All finished” will be displayed, and a directory named “**4MRNA-Results**” will be created inside the 4MRNA execution directory (i.e. the directory you created at the beginning).



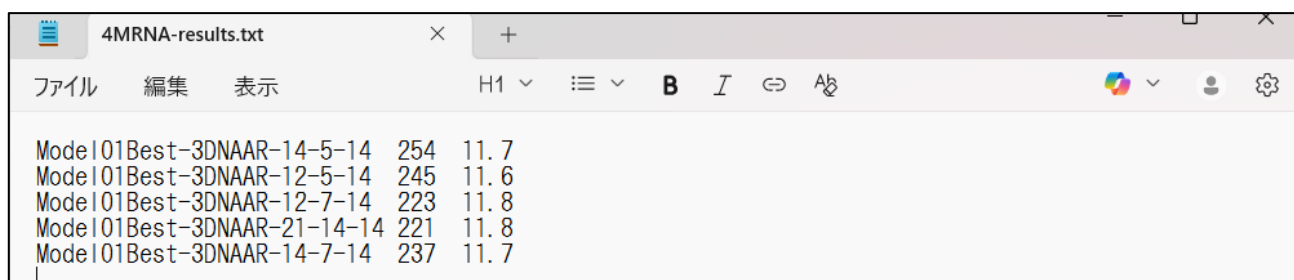
- Inside the “4MRNA-Results” directory, there is a file named “**4MRNA-results.txt**” and up to seven subdirectories. Each subdirectory name begins with “**phaser-**” and contains the results of molecular replacement (MR).



- Inside each “phaser-” subdirectory, there is a structural file named “**PHASER.1.pdb**.” This file is a solution of molecular replacement, that is, a candidate for the correct structure.
- The following figure is an example of the final directory structure.

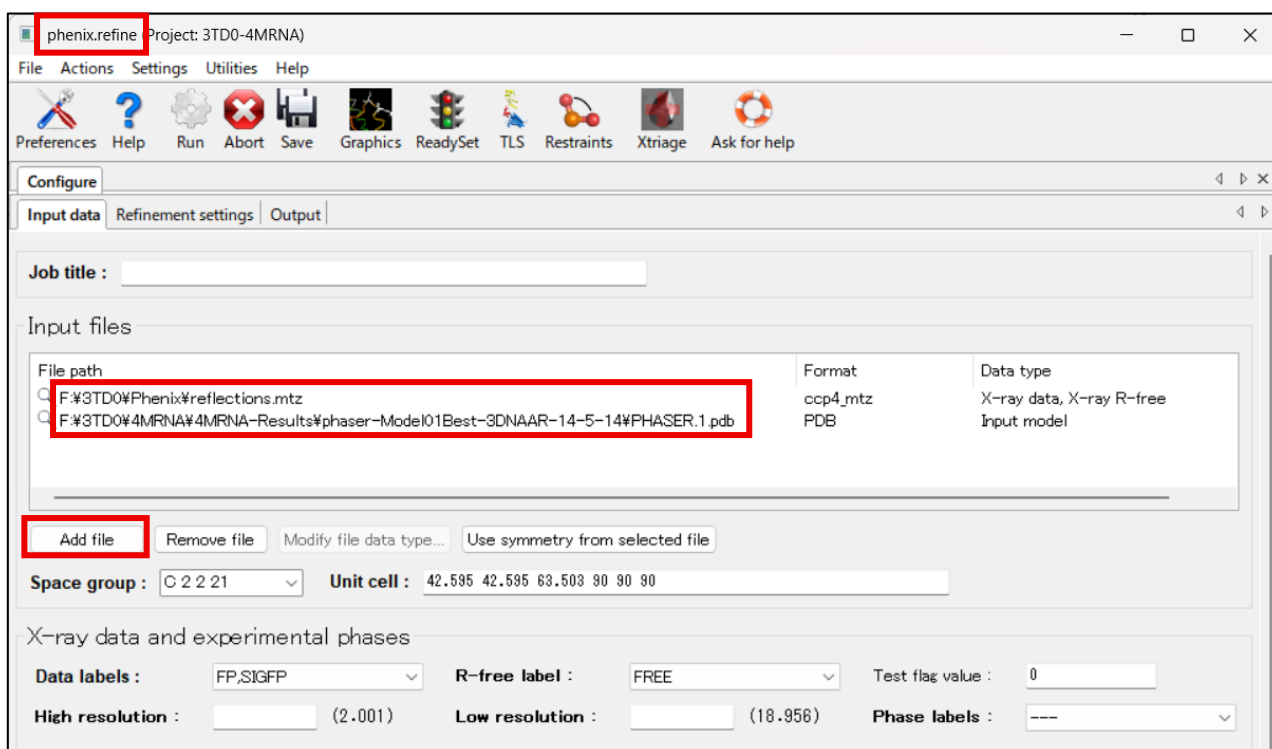


- Each run of molecular replacement produces a solution, but it must be validated against the experimental data (electron density) to confirm its correctness. This validation is guided by two statistical values, **LLG** and **TFZ**. The program is designed to automatically select up to seven solutions with the most favorable statistics. These statistics are compiled in “**4MRNA-results.txt**.” Each line in this file lists the ID of molecular replacement (i.e. the name of directory) , LLG, and TFZ in that order.



Model	LLG	TFZ
Model01Best-3DNAAR-14-5-14	254	11.7
Model01Best-3DNAAR-12-5-14	245	11.6
Model01Best-3DNAAR-12-7-14	223	11.8
Model01Best-3DNAAR-21-14-14	221	11.8
Model01Best-3DNAAR-14-7-14	237	11.7

- Based on our experience, for nucleic acid structures, solutions with LLG values greater than 100 and TFZ values greater than 8 may indicate a possible MR solution. Using these values as practical criteria, you should visually inspect the agreement between the electron density map and the model, starting from the solutions with better statistics. This can be done by loading the structure file (“**PHASER.1.pdb**”) and the electron density map (“**PHASER.1.mtz**”) into software such as Coot.
 - In some cases, multiple solutions may show good statistics exceeding the above criteria and reasonable agreement with the electron density map. In such cases, our recommendation is to select the solution with the highest TFZ value first. If multiple solutions have the same TFZ value, the solution with the highest LLG value among them should be chosen.
 - After the optimal structure has been determined, it will be used as the initial structure for refinement.
- [e.g.]** For example, if the structure with the molecular replacement ID “Model01Best-3DNAAR-14-5-14” (i.e., “PHASER.1.pdb” within the directory “phaser-Model01Best-3DNAAR-14-5-14”) is decided as the initial structure, you should add this file. In addition, add the reflection file (“reflections.mtz”).



phenix.refine Project: 3TD0-4MRNA

File Actions Settings Utilities Help

Preferences Help Run Abort Save Graphics ReadySet TLS Restraints Xtriage Ask for help

Configure

Input data Refinement settings Output

Job title :

Input files

File path	Format	Data type
F:\3TD0\Phenix\reflections.mtz	ccp4_mtz	X-ray data, X-ray R-free
F:\3TD0\4MRNA\4MRNA-Results\phaser-Model01Best-3DNAAR-14-5-14\PHASER.1.pdb	PDB	Input model

Add file Remove file Modify file data type... Use symmetry from selected file

Space group : C 2 2 21 Unit cell : 42.595 42.595 63.503 90 90 90

X-ray data and experimental phases

Data labels : FP.SIGFP R-free label : FREE Test flag value : 0

High resolution : (2.001) Low resolution : (18.956) Phase labels : ---

3-3. Supplementary notes

- Actually, for the **4MRNA** command to work, an input file called “**4MRNA-INP.txt**” is required.
- There are two ways to prepare the input file “4MRNA-INP.txt”:
 1. Prepare the input file yourself before running the **4MRNA** command.
 - ⇒ Please refer to the template below.
 2. **[Strongly recommended]** Do not prepare the input file in advance. Instead, answer the questions prompted by the **4MRNA** command, and the program will automatically create the input file.
 - ⇒ Please refer to [Section 3-2](#).
- The template of the input file “4MRNA-INP.txt” looks like the screenshot below. The following three pieces of information should be repeated for each model used in molecular replacement.
 - TYPE**: This is the nucleic acid type. Choose from A-form DNA, B-form DNA, or A-form RNA.
 - SEQ**: This is the sequence. Only the sequence of one strand of the duplex is required. The complementary strand is processed automatically by the program.
 - 5PHOS**: Choose whether to keep or remove the 5'-terminal phosphate groups.
 - NUM**: This means how many copies of the model to be searched in molecular replacement.

```
-TYPE1 A-DNA
-SEQ1 ACGTACGTACGT
-5PHOS1 Keep
-NUM1 1

-TYPE2 B-DNA
-SEQ2 ACGTACGT
-5PHOS2 Remove
-NUM2 1

-TYPE3 A-RNA
-SEQ3 ACGU
-5PHOS3 Keep
-NUM3 1
```

4. How to use the **customize mode**

In this mode, after executing the **4MRNA** command, the 4MRNA workflow does not start automatically as it does in the default mode. Instead, the programming code that runs in the background is downloaded. You can then edit and customize the code.

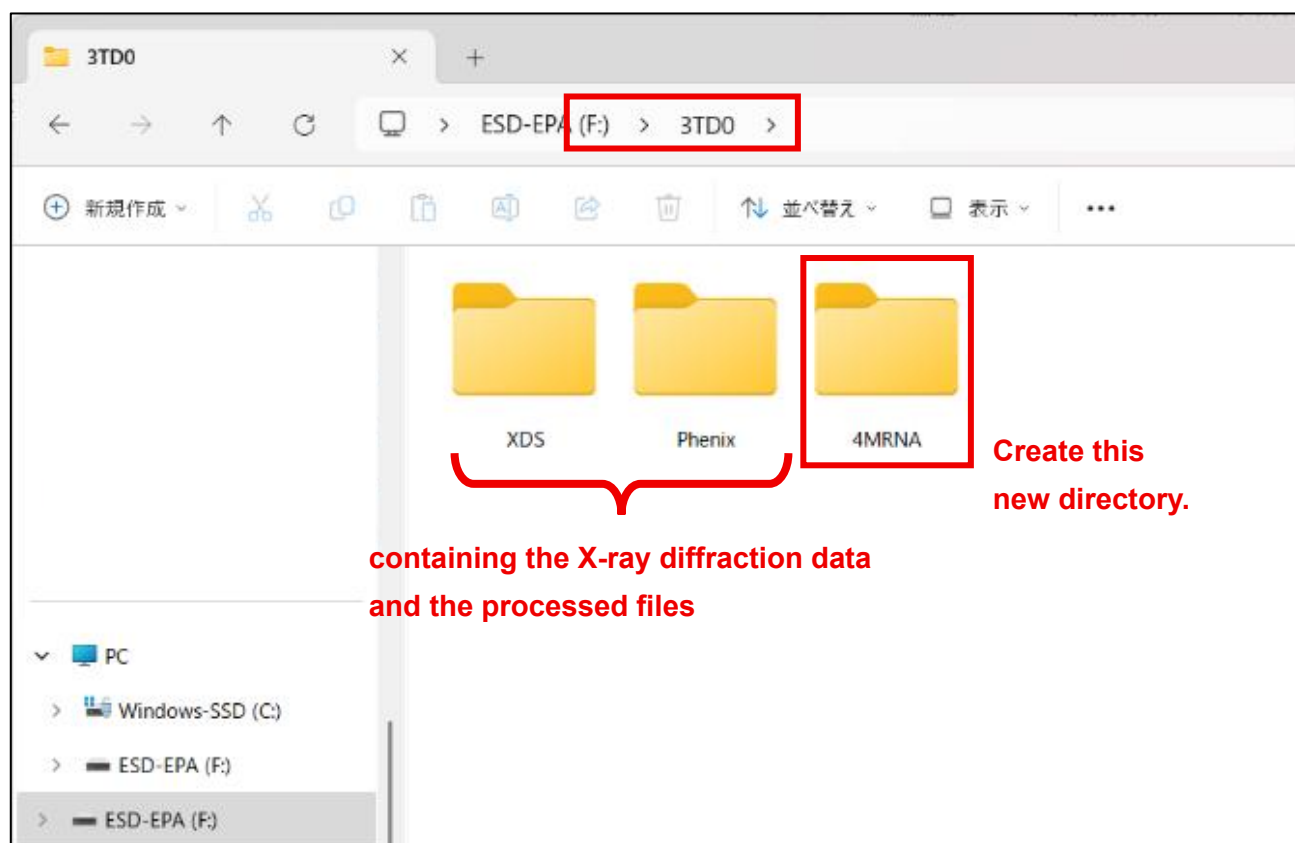
4-1. Summary of the procedure

- Step 1. Create a new directory on your computer for executing 4MRNA. After creating it, place your reflection file (.mtz) for molecular replacement into that directory.
- Step 2. Open Ubuntu/Terminal, change to the directory created in Step 1, and run the command **4MRNA**.
- Step 3. When prompted to choose the execution mode, select "**customize**".
- Step 4. You can edit each downloaded code.
- Step 5. Please run the edited scripts with the **bash** command on Ubuntu/Terminal in the order indicated by the number at the beginning of each file.
- Step 6. When 4MRNA finishes, a directory named "**4MRNA-Results**" will be created, containing up to seven candidate solutions for molecular replacement.

4-2. Step-by-step instructions with screenshots

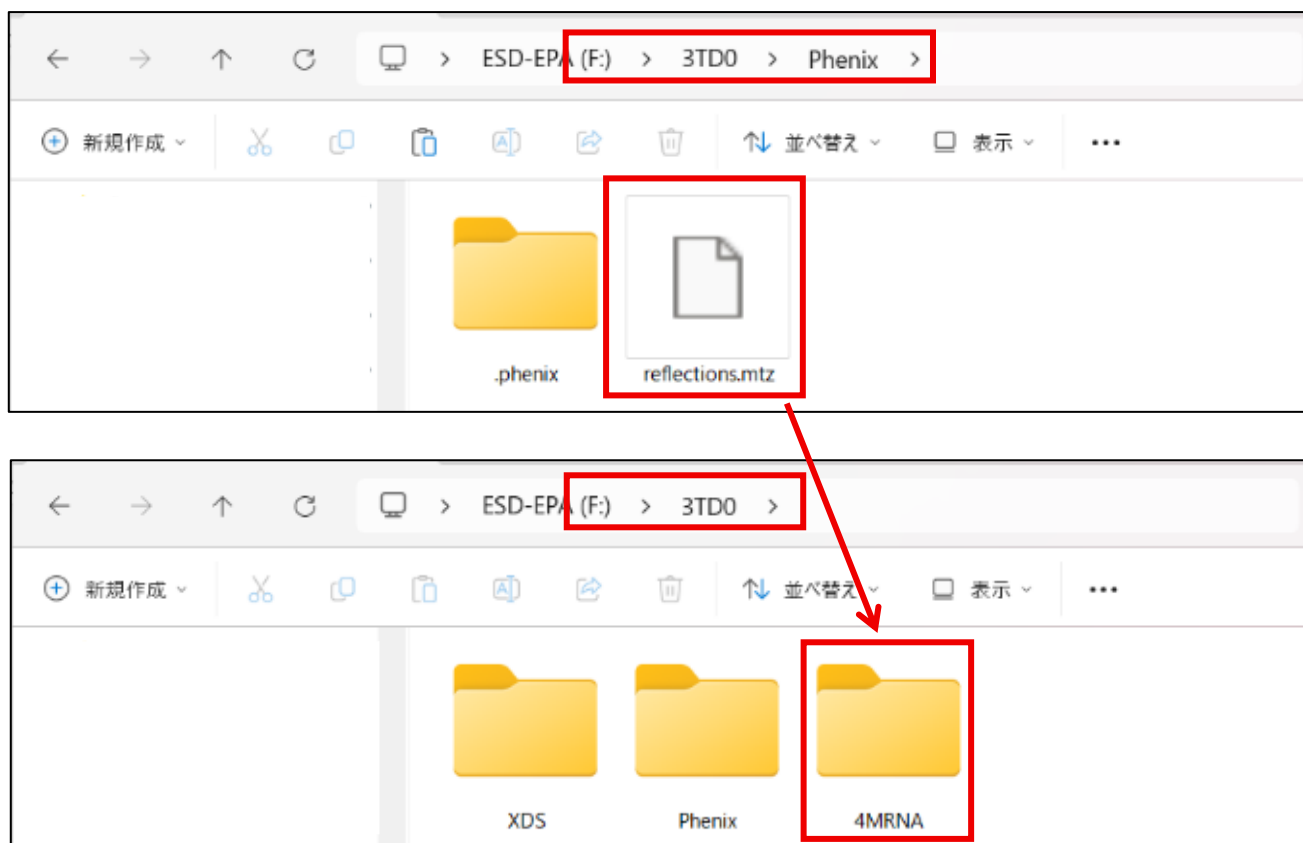
- Create a new directory on your computer for executing 4MRNA.

[e.g.] Suppose your sample is named 3TD0, and the X-ray diffraction data is stored in a directory 3TD0 on a USB drive (F:). In that case, create a new directory named "4MRNA" inside "F:\3TD0".

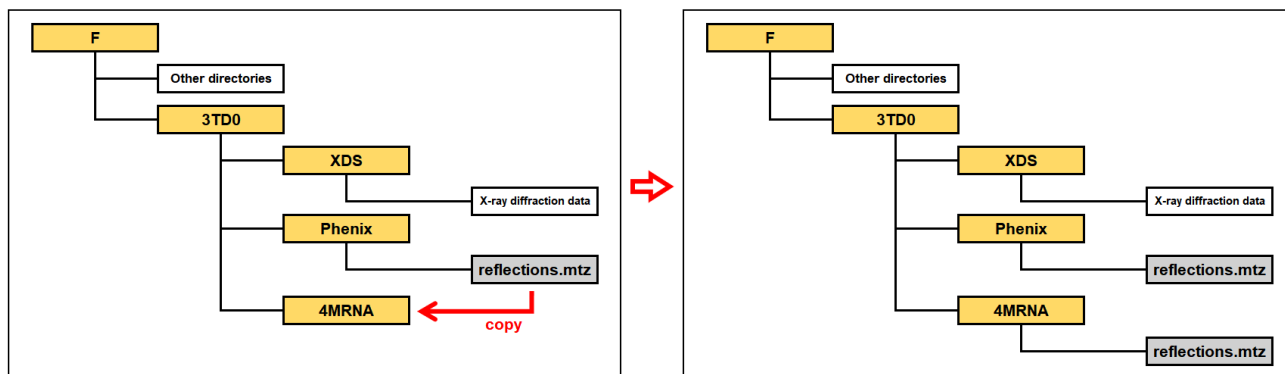


■ Place your reflection file (.mtz) for molecular replacement into the created directory.

[e.g.] When using the software Phenix, the reflection file is typically named “reflections.mtz”. If this file is in “F:\3TD0\Phenix”, copy it, and paste to “F:\3TD0\4MRNA”.



■ The directory structure after these operations is as illustrated in the example below.



■ Open Ubuntu or Terminal, change to the created directory, and run the command **4MRNA** as follows.

[e.g.] If the created directory is “F:\3TD0\4MRNA”, enter the command in Ubuntu as shown in the figure below. (This example is for Windows.)

```
shin@Shin-PC-Lenovo: /mnt/f. x + v
(base) shin@Shin-PC-Lenovo:~$ cd /mnt/f/3TD0/4MRNA
(base) shin@Shin-PC-Lenovo:/mnt/f/3TD0/4MRNA$ 4MRNA|
```


- Actually, for the command **4MRNA** to work, an input file called “**4MRNA-INP.txt**” is required. The user is prompted to choose from two options: (1) preparing the file yourself (a template will be provided), or (2) answering questions on the screen so that the program automatically generates it.
- Option (2) is strongly recommended. For option (1), refer to the supplementary notes in [Section 3-3](#).
- If you choose option (2), several questions will be asked. Please answer them. Once all questions have been answered, the 4MRNA workflow will start automatically.

[e.g.] The figure below shows an example of molecular replacement using two different RNAs as models.

In this example, during molecular replacement, one copy of the first model and two copies of the second model are searched for in the asymmetric unit of the crystal. In addition, the 5' end of a nucleic acid may retain a phosphate group or lack a phosphate group depending on the synthesis method. To allow a more accurate model to be used for molecular replacement, this can be specified in the model settings. (In this example, “Keep” option is selected with the refinement step after molecular replacement in mind.) This is a relatively minor detail and is often not critical. Therefore, if you are unsure, selecting “Keep” is generally a safe choice.

```
Do you want to run 4MRNA (v2.5.0) now? [Y/n] Y
[4MRNA] Information written to: /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Information.txt
[4MRNA] Downloading release v2.5.0...
Input file '4MRNA-INP.txt' not found.
Choose one of the following:
  1) I will prepare it myself. (A template will be provided.)
  2) I will answer questions here to generate it now.
Select [1/2]: 2
The number of models for molecular replacement [1,2,3]: 2

Enter information of model #1.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): CGUCGACG
Do you want to keep or remove the 5'-terminal phosphate group? [Keep,Remove]: Keep
How many copies of the model to be searched in molecular replacement: 1

Confirm model #1:
-TYPE1 A-RNA
-SEQ1 CGUCGACG
-SPHOS1 Keep
-NUM1 1
Is this correct? [Y/N]: Y

Enter information of model #2.
Nucleic acid type [A-DNA,B-DNA,A-RNA]: A-RNA
Sequence (one strand of duplex, 5'→3'): GCGUC
Do you want to keep or remove the 5'-terminal phosphate group? [Keep,Remove]: Keep
How many copies of the model to be searched in molecular replacement: 2

Confirm model #2:
-TYPE2 A-RNA
-SEQ2 GCGUC
-SPHOS2 Keep
-NUM2 2
Is this correct? [Y/N]: Y
```

- When prompted with “Please choose execution mode.”, please enter “**customize**”.

```
4MRNA-INP.txt created
-TYPE1 A-RNA
-SEQ1 CGUCGACG
-SPHOS1 Keep
-NUM1 1

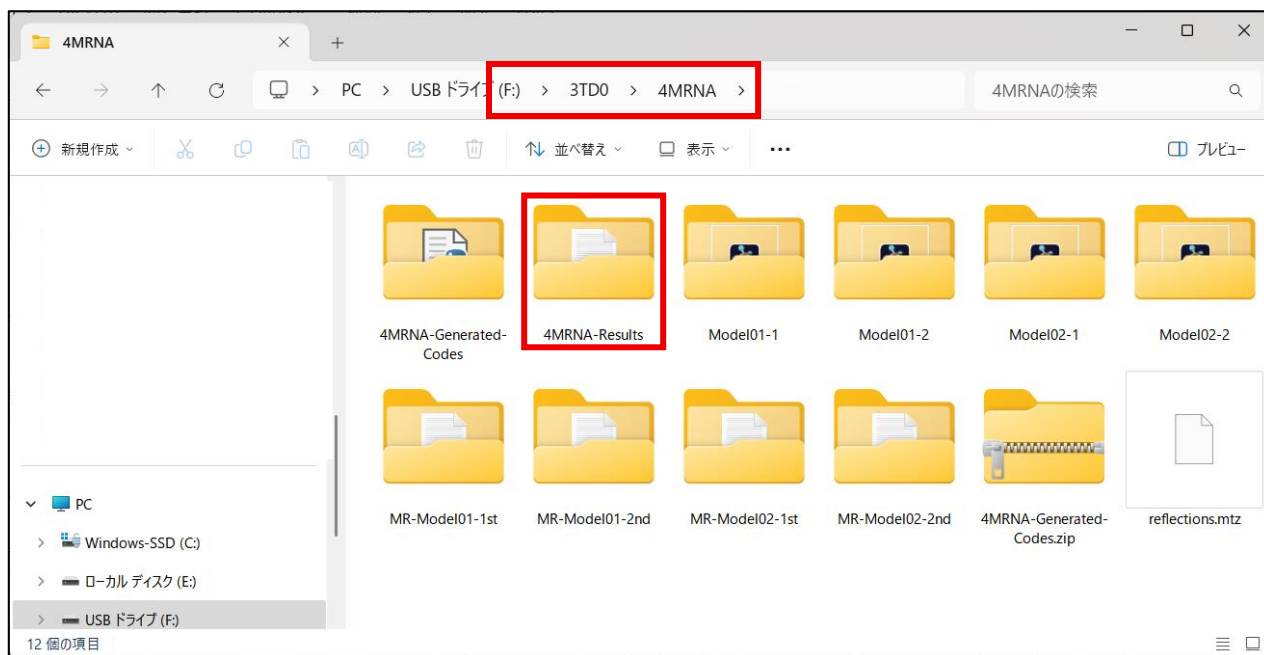
-TYPE2 A-RNA
-SEQ2 GCGUC
-SPHOS2 Keep
-NUM2 2

Continuing...
Detected model types: 2
Model01: TYPE=A-RNA SEQ=CGUCGACG SPHOS=Keep NUM=1 MW=5051.38
Model02: TYPE=A-RNA SEQ=GCGUC SPHOS=Keep NUM=2 MW=3115.14
Downloading: 01.Shell-CreatingModels-1st.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/01.Shell-CreatingModels-1st.sh
Downloading: 02.Shell-MR-2model-Model01-1st.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/02.Shell-MR-2model-Model01-1st.sh
Downloading: 03.Shell-CreatingModels-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/03.Shell-CreatingModels-2nd.sh
Downloading: 04.Shell-MR-2model-Model01-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/04.Shell-MR-2model-Model01-2nd.sh
Downloading: 05.Shell-MR-2model-Model02-1st.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/05.Shell-MR-2model-Model02-1st.sh
Downloading: 06.Shell-CreatingModels-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/06.Shell-CreatingModels-2nd.sh
Downloading: 07.Shell-MR-2model-Model02-2nd.sh -> /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/07.Shell-MR-2model-Model02-2nd.sh
Downloading: min.params -> /mnt/h/4MRNA-Update/v2.5.0/test3/min.params
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/01.Shell-CreatingModels-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/03.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/06.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/01.Shell-CreatingModels-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/02.Shell-MR-2model-Model01-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/03.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/04.Shell-MR-2model-Model01-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/05.Shell-MR-2model-Model02-1st.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/06.Shell-CreatingModels-2nd.sh
Updated /mnt/h/4MRNA-Update/v2.5.0/test3/4MRNA-Background-Codes/tmp/run.5R825x/4MRNA-Background-Codes/07.Shell-MR-2model-Model02-2nd.sh
Please choose execution mode. [default,customize]: customize
```

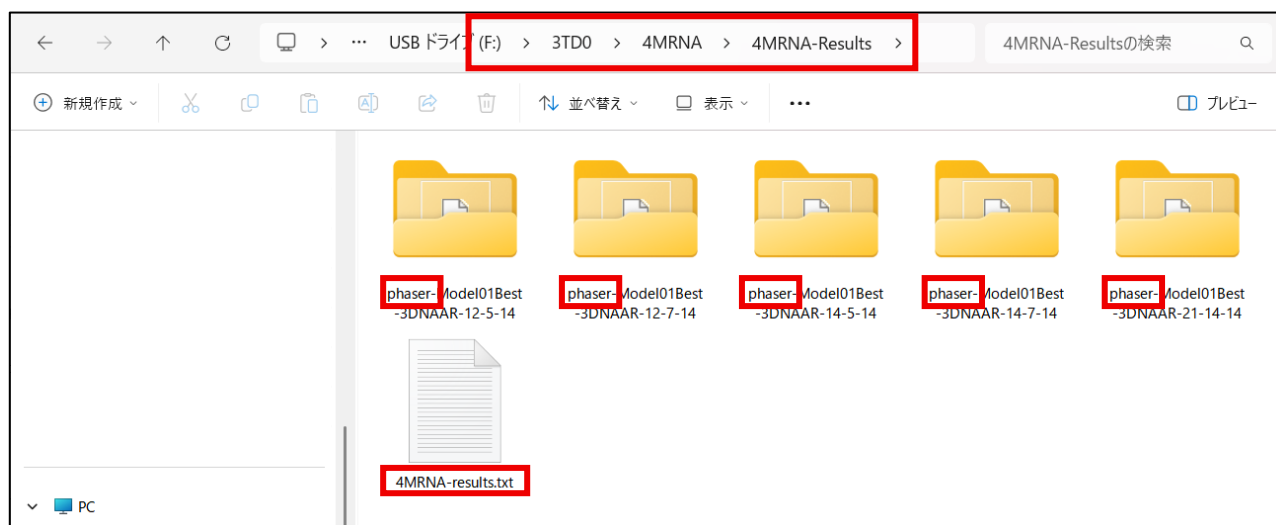
- The codes running in the background of 4MRNA are downloaded as a folder named “**4MRNA-Background-Codes**”.
- Basically, you only need to run the codes in the order indicated by the numbers at the beginning of each file, and the entire 4MRNA workflow will be completed. Please open Ubuntu or Terminal, change to the directory “4MRNA-Background-Codes”, and execute the code as follows.

```
bash <file name>.sh
```

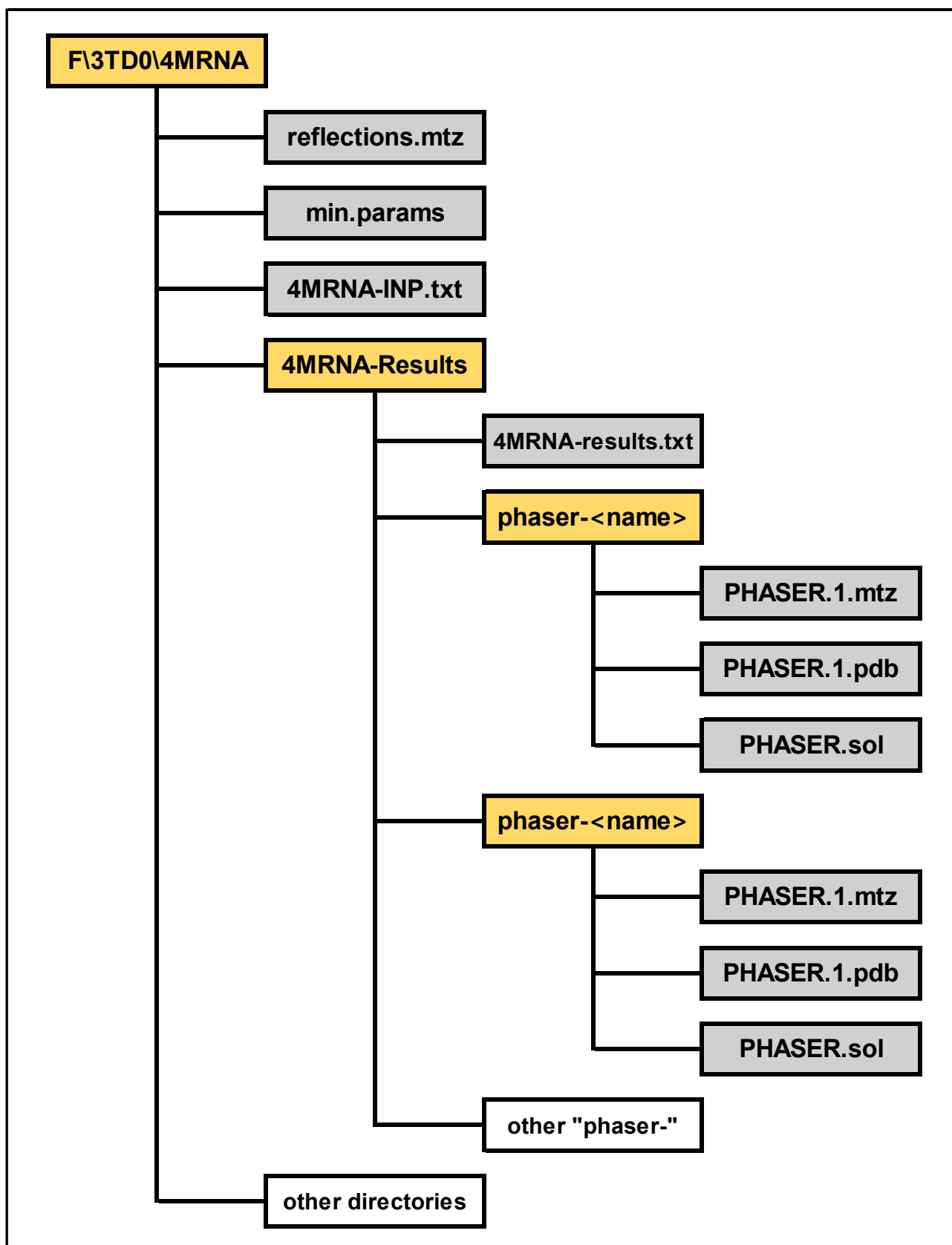
- However, it is also possible to review and modify the contents of each file if necessary.
- After all the codes have finished running, a directory named “**4MRNA-Results**” will be created.



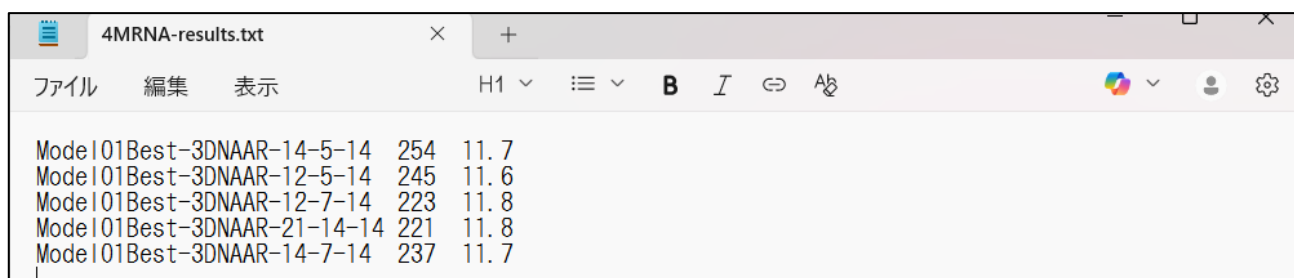
- Inside the “4MRNA-Results” directory, there is a file named “**4MRNA-results.txt**” and up to seven subdirectories. Each subdirectory name begins with “**phaser-**” and contains the results of molecular replacement (MR).



- Inside each “phaser-” subdirectory, there is a structural file named “**PHASER.1.pdb**.” This file is a solution of molecular replacement, that is, a candidate for the correct structure.
- The following figure is an example of the final directory structure.

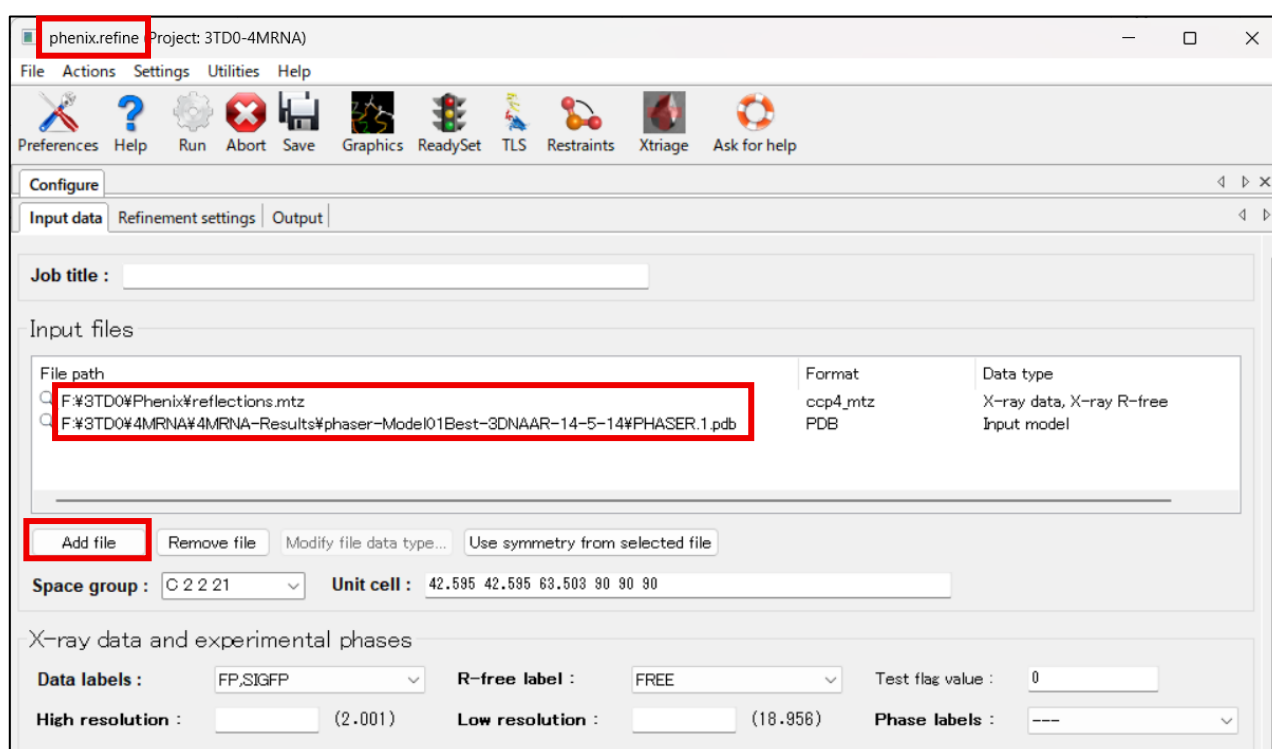


- Each run of molecular replacement produces a solution, but it must be validated against the experimental data (electron density) to confirm its correctness. This validation is guided by two statistical values, **LLG** and **TFZ**. The program is designed to automatically select up to seven solutions with the most favorable statistics. These statistics are compiled in “**4MRNA-results.txt**.” Each line in this file lists the ID of molecular replacement (i.e. the name of directory) , LLG, and TFZ in that order.



Model	LLG	TFZ
Model01Best-3DNAAR-14-5-14	254	11.7
Model01Best-3DNAAR-12-5-14	245	11.6
Model01Best-3DNAAR-12-7-14	223	11.8
Model01Best-3DNAAR-21-14-14	221	11.8
Model01Best-3DNAAR-14-7-14	237	11.7

- Based on our experience, for nucleic acid structures, solutions with LLG values greater than 100 and TFZ values greater than 8 may indicate a possible MR solution. Using these values as practical criteria, you should visually inspect the agreement between the electron density map and the model, starting from the solutions with better statistics. This can be done by loading the structure file (“**PHASER.1.pdb**”) and the electron density map (“**PHASER.1.mtz**”) into software such as Coot.
 - In some cases, multiple solutions may show good statistics exceeding the above criteria and reasonable agreement with the electron density map. In such cases, our recommendation is to select the solution with the highest TFZ value first. If multiple solutions have the same TFZ value, the solution with the highest LLG value among them should be chosen.
 - After the optimal structure has been determined, it will be used as the initial structure for refinement.
- [e.g.]** For example, if the structure with the molecular replacement ID “Model01Best-3DNAAR-14-5-14” (i.e., “PHASER.1.pdb” within the directory “phaser-Model01Best-3DNAAR-14-5-14”) is decided as the initial structure, you should add this file. In addition, add the reflection file (“reflections.mtz”).



phenix.refine Project: 3TD0-4MRNA

File Actions Settings Utilities Help

Preferences Help Run Abort Save Graphics ReadySet TLS Restraints Xtriage Ask for help

Configure

Input data Refinement settings Output

Job title :

Input files

File path	Format	Data type
F:\3TD0\Phenix\reflections.mtz	ccp4_mtz	X-ray data, X-ray R-free
F:\3TD0\4MRNA\4MRNA-Results\phaser-Model01Best-3DNAAR-14-5-14\PHASER.1.pdb	PDB	Input model

Add file Remove file Modify file data type... Use symmetry from selected file

Space group : C 2 2 21 Unit cell : 42.595 42.595 63.503 90 90 90

X-ray data and experimental phases

Data labels : FP.SIGFP R-free label : FREE Test flag value : 0

High resolution : (2.001) Low resolution : (18.956) Phase labels : ---

5. Notes

5-1. Citation

If you use 4MRNA in your research, please cite the following reference.

Ando, S., & Kondo, J. (2026). 4MRNA: a new approach for nucleic acid molecular replacement using models with diverse parameter patterns. *RNA*, 32(10), 1544–1559.

<https://doi.org/10.1261/rna.081112.126>

In addition, please also cite the external software packages used by 4MRNA.

- **3DNA:** Lu, X. J., & Olson, W. K. (2008). 3DNA: a versatile, integrated software system for the analysis, rebuilding and visualization of three-dimensional nucleic-acid structures. *Nature protocols*, 3(7), 1213–1227. <https://doi.org/10.1038/nprot.2008.104>
- **Phenix:** Liebschner, D., Afonine, P. V., Baker, M. L., *et al.* (2019). Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix. *Acta Crystallographica Section D: Structural Biology*, 75(Pt 10), 861–877. <https://doi.org/10.1107/S2059798319011471>
- **Phaser:** McCoy, A. J., Grosse-Kunstleve, R. W., Adams, P. D., Winn, M. D., Storoni, L. C., & Read, R. J. (2007). Phaser crystallographic software. *Journal of applied crystallography*, 40(Pt 4), 658–674. <https://doi.org/10.1107/S0021889807021206>

5-2. License information

- **4MRNA itself is distributed under the [MIT License](#).** This means that it can be used completely freely, provided that the copyright notice is included.
- However, please pay attention to the licenses of the dependencies. 4MRNA relies on several external resources, including 3DNA, Phenix, and CCP4/Phaser, but these components are not included in the distribution of 4MRNA. Users are responsible for installing them separately and complying with their respective licenses.
- A summary of these dependencies is provided in “[DEPENDENCIES.txt](#)” on GitHub.

5-3. Supported environment

4MRNA can run in any environment that supports shell scripts and the external software packages 3DNA and Phaser. It has been tested on the following operating systems.

- Windows (via Windows Subsystem for Linux, Ubuntu)
- macOS
- Linux (Rocky Linux)